

Q. Write a program in C++ to sort the numbers in an array using separate functions for read, display, sort and swap. The objective of this assignment is to learn the concepts of input, output, functions, call by reference in C++.

```
#include <iostream>
using namespace std;

// Function declarations
void inputArray(int arr[], int size);
void displayArray(int arr[], int size);
void sortArray(int arr[], int size);
void swapNumbers(int &a, int &b); // Added swap function declaration

int main() {
    int arr[2], i;
    int size = sizeof(arr) / sizeof(arr[0]);

    cout << "\nNumber of elements in array: " << size << endl;

    inputArray(arr, size);

    cout << "\nThe elements in the array are: ";
    displayArray(arr, size);

    // First sort the array
    sortArray(arr, size);
    cout << "\nThe elements in the array after sorting are: ";
    displayArray(arr, size);

    if (size == 2) {
        swapNumbers(arr[0], arr[1]);
    }
    cout << "\nThe elements in the array after swap are: ";
    displayArray(arr, size);

    return 0;
}
```

```
void inputArray(int arr[], int size) {  
    cout << "\nEnter " << size << " integers: " << endl;  
    for (int i = 0; i < size; i++) {  
        cin >> arr[i];  
    }  
}
```

```
void displayArray(int arr[], int size) {  
    for (int i = 0; i < size; i++) {  
        cout << arr[i] << " ";  
    }  
    cout << endl;  
}
```

```
void sortArray(int arr[], int size) {  
    if (size == 2 && arr[0] > arr[1]) {  
        swapNumbers(arr[0], arr[1]);  
    }  
}
```

```
void swapNumbers(int &a, int &b) {  
    int temp = a;  
    a = b;  
    b = temp;  
}
```

Number of elements in array: 2

Enter 2 integers:

2 8

The elements in the array are: 2 8

The elements in the array after sorting are: 2 8

The elements in the array after swap are: 8 2

=== Code Execution Successful ===